

Maximising Edge Density in Graphs Subject to Connectivity Constraints

Erdős Problem 915, from Proof to Search

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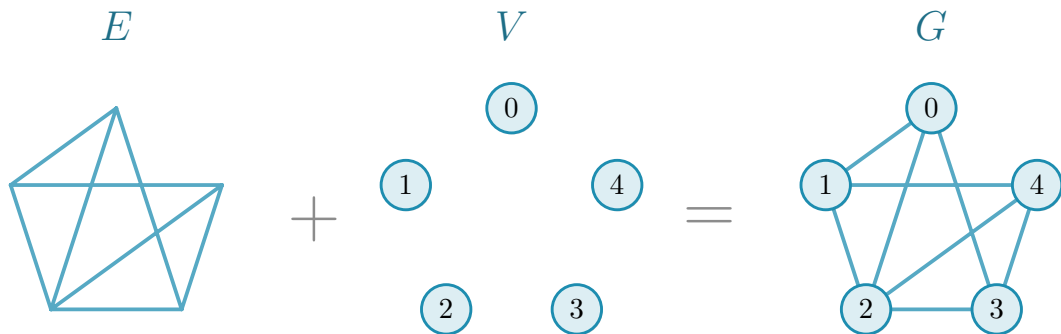
Maximising edge density in graphs
under connectivity constraints



translation

networks with **many links** and **low connectivity**

Graph

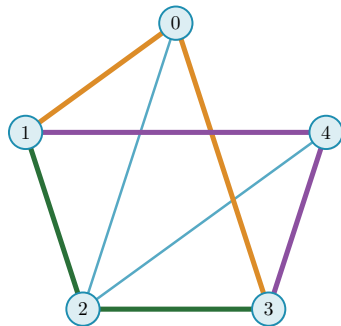


Erdős Problem 915

$$\ell_m(n) = \max_{|V(G)|=n} \left\{ |E(G)| : \lambda_G^{\max} \leq m-1 \right\}$$

$$\lambda_G^{\max} = \max_{u \neq v} \lambda_G(u, v)$$

$\lambda_G(u, v)$ = the largest number of $u-v$ routes that share **no edge**: one colour each

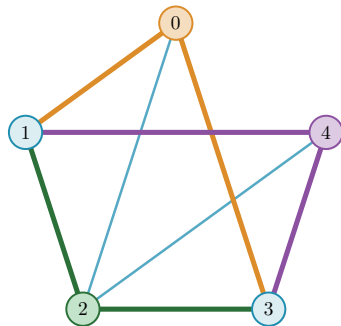


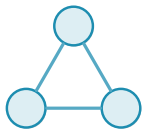
The same question, vertex by vertex

$$k_m(n) = \max_{|V(G)|=n} \left\{ |E(G)| : \kappa_G^{\max} \leq m-1 \right\}$$

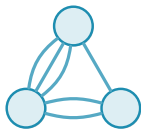
$$\kappa_G^{\max} = \max_{u \neq v} \kappa_G(u, v)$$

$\kappa_G(u, v)$ = the largest number of u - v routes that share **no interior vertex**

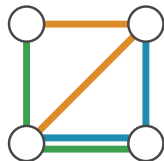




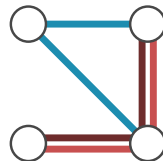
simple graph



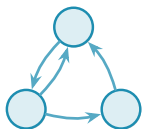
multigraph



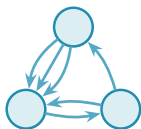
hypergraph



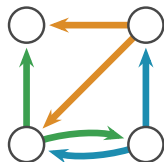
multihypergraph



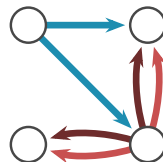
simple digraph



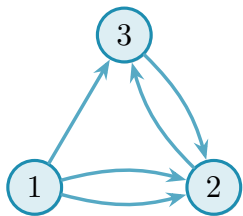
directed multigraph



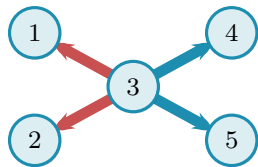
directed hypergraph



directed multihypergraph



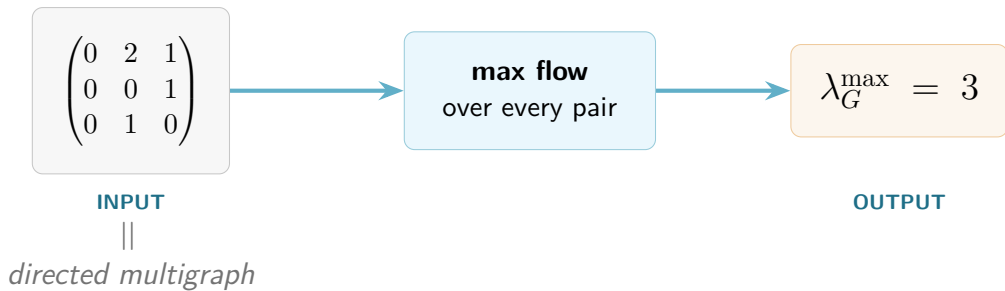
$$\begin{pmatrix} \mathbf{0} & 2 & 1 \\ 0 & \mathbf{0} & 1 \\ 0 & 1 & \mathbf{0} \end{pmatrix}$$



list of hyperedges:

$$e_1 : \{3 \mid 1, 2\}$$

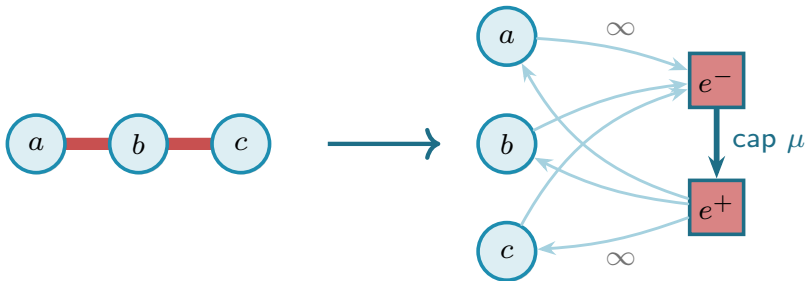
$$e_2 : \{3 \mid 4, 5\}$$



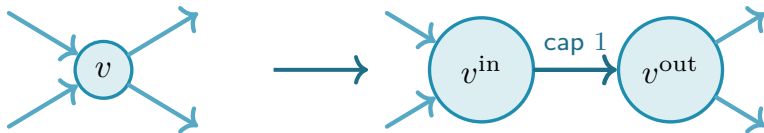
undirected $\rightarrow$ directed



hypergraph $\rightarrow$ non-hypergraph



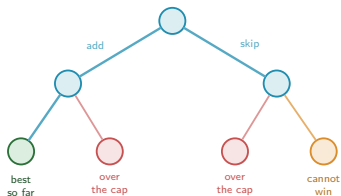
$$\kappa \rightarrow \lambda$$



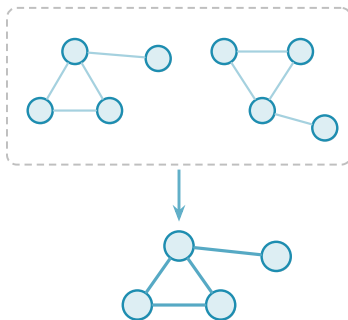
every intermediate vertex is split this way
the flow itself runs from s^{out} to t^{in}

Efficiency: three ways to make the sweep finish

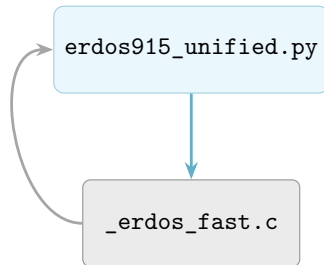
pruning



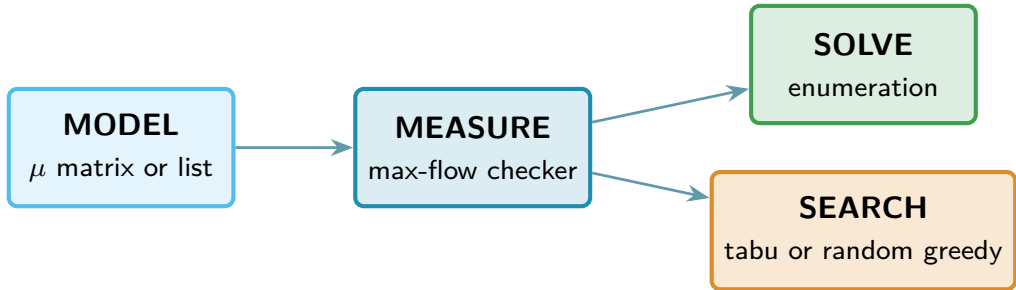
generating search



C: faster max flow

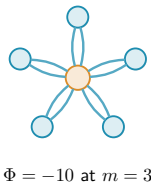
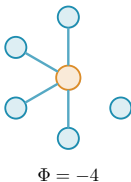
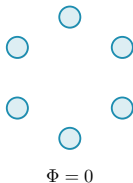


Solving versus searching



The search: one number per graph

$$\Phi(G) = \underbrace{-|E(G)|}_{\substack{\text{reward} \\ \text{more edges, lower energy}}} + \underbrace{\Lambda \cdot \max(0, c^{\max}(G) - (m - 1))}_{\substack{\text{penalty} \\ \text{discourages excess connectivity}}}$$

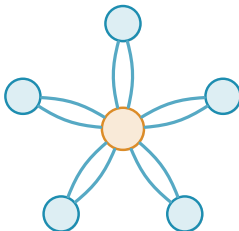


Extremal families the search rediscovers



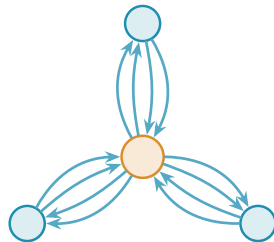
tree

$$n = 6, \ell_2(n) = 5$$



multigraph star

$$n = 6, m = 3, L_3(n) = 10$$

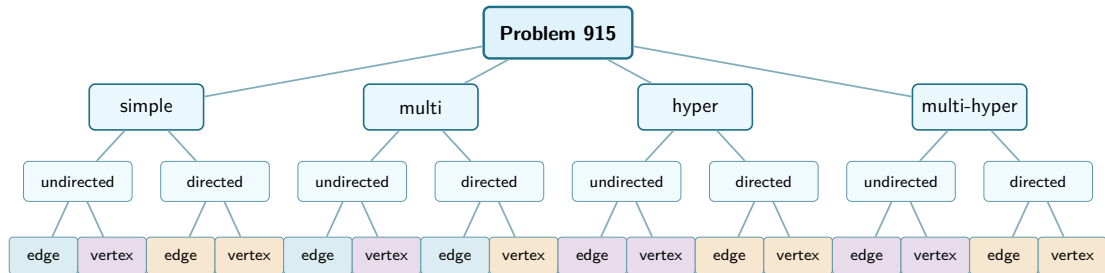


double star

$$n = 4, m = 3, L_3^{\text{dir}}(n) = 12$$

examples of the families, not the saved search witnesses

Status of the sixteen variants



exact value established in the graph domain



general exact value open, bounds and small cases known



partial results or a conjectural general formula

Thank you

`github.com/louisvandenbruwaene/thesis`

Backup: the sixteen variants

WHAT IS KNOWN ABOUT A BOUND

PROVED

established, with a proof the author has checked

PROPOSED

conjecture with a proposed proof the author has not checked line by line

OPEN

no general value established

WHAT A CONSTRUCTION IS WORTH

EXTREMAL

attains the maximum in the stated range

NEAR-EXTREMAL

the densest family reached here, with no matching upper bound

ASYMPTOTICALLY EXTREMAL

matches the leading term of the upper bound

Simple graph · edge

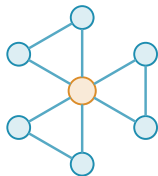
$$\ell_m(n)$$

UPPER $\left\lfloor \frac{m(n-1)}{2} \right\rfloor$ **PROVED** Mader, via Gomory–Hu

LOWER $\left\lfloor \frac{m(n-1)}{2} \right\rfloor$ **PROVED** for every $n \geq m$

- the two meet for $n \geq m$, so the value is exact there
- for $n < m$ it is $\binom{n}{2}$: not even K_n carries m routes

COMPUTED Tabu search returns the tree at $n = 6, m = 2$: $5 = \ell_2(6)$.



hub-and-spokes **EXTREMAL**

Simple graph · vertex

$$k_m(n)$$

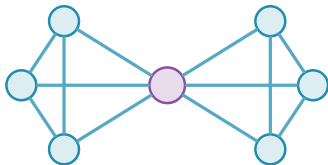
UPPER $k_m(n) = \ell_m(n)$ for $m \leq 4$, $k_5(n) = \lfloor \frac{8n}{3} \rfloor - 4$ for $n \geq 14$ **PROVED**

$m \geq 6$: **OPEN**

LOWER $c_m > m/2$ for $m \geq 5$ **PROPOSED** under the block formulation

- at $m = 5$ the two agree to $n = 13$, first gap 33 against 32 at $n = 14$
- $c_m = \lim_n k_m(n)/(n - 1)$, and $m/2 < c_m \leq 2(m - 1)$ for $m \geq 5$

COMPUTED Block sweeps reach small blocks only, and the recursive constructions in the literature show why they cannot fix c_m .



2-connected blocks at a cut vertex **NEAR-EXTREMAL**

Simple digraph · arc

$$\ell_m^{\text{dir}}(n)$$

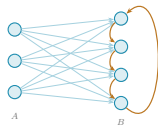
UPPER $M(n) = \max\left(2(n-1), \left\lfloor \frac{n^2}{4} \right\rfloor\right)$ at $m = 2$ **PROVED**

$\left\lfloor \frac{n^2}{4} \right\rfloor + 4(m-1)(n-1)$ at every n **PROVED**

LOWER $\frac{n^2}{4} + \frac{(m-2)n}{2} + O_m(1)$ **PROVED**

- for fixed $m \geq 3$, $\ell_m^{\text{dir}}(n) = n^2/4 + \Theta_m(n)$; the linear coefficient is open
- this family refutes the first guess $\max(m(n-1), \left\lfloor \frac{n^2}{4} \right\rfloor)$ at $m = 3$, $n = 10$

COMPUTED Enumeration confirms $m = 2$ through $n = 7$. Tabu returns 6, 10 and 12 at $n = 4$, 6 and 7.



augmented bipartite digraph

ASYMPTOTICALLY EXTREMAL

Simple digraph · vertex

$$k_m^{\text{dir}}(n)$$

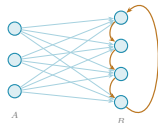
UPPER $M(n)$ at $m = 2$ **PROVED**

$$\frac{n^2}{4} + \Theta_m(n) \text{ for fixed } m \geq 3 \quad \textbf{PROVED}$$

LOWER the same bipartite family **PROVED**

- equality with the arc value at $m \geq 3$ is open: an arc upper bound does not constrain the vertex-feasible family, which contains it

COMPUTED Enumeration confirms $m = 2$ through $n = 7$, and the two separations agree in the completed $m = 3$ runs through $n = 6$.



augmented bipartite digraph

ASYMPTOTICALLY EXTREMAL

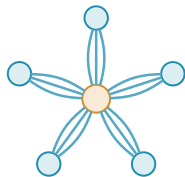
Multigraph · edge

$$L_m(n)$$

UPPER $(m-1)(n-1)$ **PROVED** a cut count, any shape

LOWER $(m-1)(n-1)$ **PROVED**

- exact for every n and m
- multiplicity removes the simple graph's distance-one restriction



tree, every edge repeated
 $m-1$ times **EXTREMAL**

COMPUTED Tabu search returns the multiplicity-2 star at $n=6$, $m=3$: $10 = L_3(6)$.

$$K_m(n)$$

UPPER $K_2(n) = n - 1$, $K_3(n) = 2(n - 1)$ **PROVED**

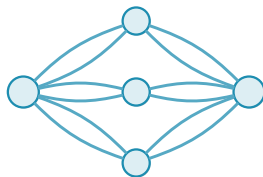
$$K_m(n) \leq (2 + o(1)) m^2 n$$
 PROPOSED

LOWER $K_3(n) \geq 2(n - 1)$ **PROVED**

$$K_m(n) \geq (3 - 2\sqrt{2} + o(1)) m^2 n$$
 PROPOSED

- the two coefficients bracket the growth as $m \rightarrow \infty$ with $n/m \rightarrow \infty$, and no common limit is established
- parallel copies are separated routes, so this is not the simple problem: $K_5(4) = 14$ against $L_5(4) = 12$

COMPUTED $K_5(4) = 14$ and $K_5(5) = 19$ by exhaustion. A nine-vertex sweep over all 191826 two-connected graphs with at least fourteen edges gives $g_6(9) = 54$.



thickened complete bipartite
block **NEAR-EXTREMAL**

Directed multigraph · arc

$$L_m^{\text{dir}}(n)$$

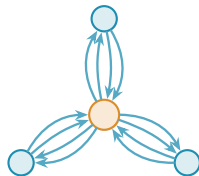
UPPER $(m-1)M(n)$ **PROVED** a sparse skeleton per unit of $m-1$

LOWER $(m-1)M(n)$ **PROVED** two branches, both attained

- exact for every n and m , and the fractional optimum is $M(n)$ too, so one m -free optimisation bounds every threshold
- the classification of the extremisers for $n \geq \max(8, 2m)$ stays

PROPOSED

COMPUTED Tabu returns the double star at $m=3$: 12 at $n=4$ and 16 at $n=5$.



thickened bidirected tree

EXTREMAL

linear branch, $n \leq 7$

Directed multigraph · vertex

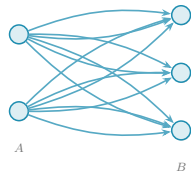
$$K_m^{\text{dir}}(n)$$

UPPER $(m-1)\frac{n^2}{4} + O_m(n)$ **PROVED**

LOWER $(m-1)\lfloor n^2/4 \rfloor$, thickened bipartite **PROVED**

- the arc extremiser is not even a finite optimum here:
 $K_6^{\text{dir}}(3) = 24$ against an established arc maximum of 20
- the exact correction and the best block shapes are open

COMPUTED Direct enumeration gives $K_6^{\text{dir}}(3) = 24$.



thickened one-way bipartite
digraph

ASYMPTOTICALLY EXTREMAL

Simple hypergraph · edge

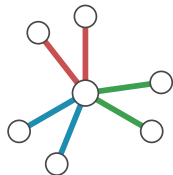
$$\ell_m^{(r)}(n)$$

UPPER $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$ **PROVED** induction on a hyperedge cut

LOWER the same value **PROVED** when $m-1 \leq \binom{n-2}{r-2}$

- one hyperedge spans r vertices but serves only one route
- outside that range attainment is not guaranteed: at $n = r = m = 3$ the bound reads 2 and only one hyperedge exists

COMPUTED Exhaustive enumeration gives 11 at $(n, m, r) = (6, 6, 3)$, below the bound 12.



star hypertree ($m = 2$)

EXTREMAL

Simple hypergraph · vertex

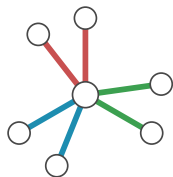
$$k_m^{(r)}(n)$$

UPPER $\left\lfloor \frac{n-1}{r-1} \right\rfloor$ at $m = 2$, $\left\lfloor \frac{2(n-1)}{r-1} \right\rfloor$ at $m = 3$ **PROVED**
 $m \geq 4$: **OPEN**

LOWER star at $m = 2$, degree-capped hub at $m = 3$ **PROVED**
 $m = 3$ for $2 < r < n$

- $m = 2$ is feasible exactly when the incidence graph is a forest
- $m = 3$ comes from the incidence-rank inequality
- an edge bound does not transfer: $\kappa \leq \lambda$ points the wrong way, vertex feasibility being the weaker restriction

COMPUTED Enumeration at $m = 3$, $r = 3$ gives 1, 3, 4 for $n = 3, 4, 5$: the bound $\lfloor 2(n-1)/(r-1) \rfloor$ from $n = 4$ on, one short at $n = r = 3$.



star hypertree ($m = 2$)

EXTREMAL

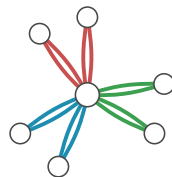
Multihypergraph · edge

$$L_m^{(r)}(n)$$

UPPER $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$ **PROVED** unchanged by repetition

LOWER $\frac{(m-1)(n-1)}{r-1}$ **PROVED** when $(r-1) \mid (n-1)$

- the constructions change even though the bound does not
- at $n = r = m = 3$, two copies of the only triple reach a value the simple model cannot



star hypertree, edges repeated
 $m-1$ times **EXTREMAL**

COMPUTED A twelve-copy construction at $(n, m, r) = (6, 6, 3)$ exceeds the simple edge optimum of 11.

Multihypergraph · vertex

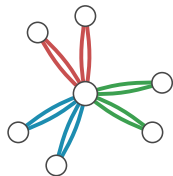
$$K_m^{(r)}(n)$$

UPPER $\left\lfloor \frac{n-1}{r-1} \right\rfloor$ at $m = 2$, $\left\lfloor \frac{2(n-1)}{r-1} \right\rfloor$ at $m = 3$ **PROVED**
larger m : **OPEN**

LOWER the repeated star adds attainment cases at $m = 3$
PROVED

- repetition does not help at $m = 2$: two copies of one hyperedge are already two separated routes

COMPUTED The twelve-copy edge witness at $(6, 6, 3)$ is a vertex lower bound, and does not prove the vertex maximum is twelve. A sweep at $m = 4$ attains $\lfloor 3(n-1)/(r-1) \rfloor$ and excludes one more in eleven multihypergraph cases, all $r \geq 3$.



repeated star hypertree

EXTREMAL $m \leq 3$, and
 $(r-1) \mid (n-1)$ at $m = 3$

Directed simple hypergraph · arc

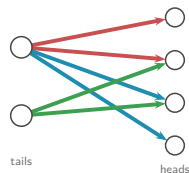
$$\ell_m^{\text{dir}(r)}(n)$$

UPPER $(1 + o(1)) \frac{(m-1)n^2}{4(r-1)}$ **PROVED** shadow digraph

LOWER the same constant **PROVED** two-layer, $r \geq 3$

- the two meet, so the leading term is settled and the finite exact values are **OPEN**
- the matching argument covers distinct two-step routes sharing one hyperedge

COMPUTED At $n = m = 6$, $r = 3$, five distinct outgoing hyperedges per tail give 30, with a direct degree-cut feasibility check.



two-layer construction

ASYMPTOTICALLY EXTREMAL

Directed simple hypergraph · vertex

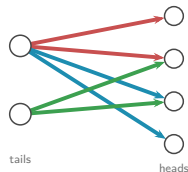
$$k_m^{\text{dir}(r)}(n)$$

UPPER $(1 + o(1)) \frac{(m-1)n^2}{4(r-1)}$ **PROVED**

LOWER the same constant **PROVED** two-layer, $r \geq 3$, fixed m and r
as $n \rightarrow \infty$

- the shadow argument re-runs under the stricter separation, so the leading term is the same
- whether the finite edge and vertex maxima agree is **OPEN**

COMPUTED Random greedy growth with shuffled restarts measures what the implemented search finds, and establishes no optimum.



two-layer construction

ASYMPTOTICALLY EXTREMAL

Directed multihypergraph · arc

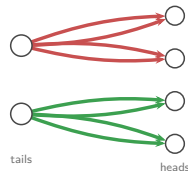
$$L_m^{\text{dir}(r)}(n)$$

UPPER $(1 + o(1)) \frac{(m-1)n^2}{4(r-1)}$ **PROVED** forward, backward and general alike

LOWER the same constant **PROVED** repeats lift the head-set limit

- the finite values can differ from the simple model, and are **OPEN**
- at $r = 2$ these are directed multigraphs, where the simple coefficient is $1/4$ instead, which is why $r \geq 3$ is kept

COMPUTED The plotted experiments use rank three and the forward convention. General orientations are checked separately in the appendix.



two-layer, hyperedges repeated

ASYMPTOTICALLY EXTREMAL

Directed multihypergraph · vertex

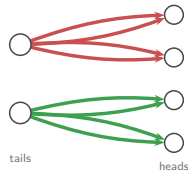
$$K_m^{\text{dir}(r)}(n)$$

UPPER $(1 + o(1)) \frac{(m-1)n^2}{4(r-1)}$ **PROVED**

LOWER the same constant **PROVED**

- the same leading term under both separations
- finite exact values **OPEN**, and whether forward and general agree once $n > r$ is open too

COMPUTED The rank-three forward-convention runs are not an exhaustive comparison of the three orientation models.



two-layer, hyperedges repeated

ASYMPTOTICALLY EXTREMAL